Fiscal Year:	FY 2005	**Task Last Updated:**	FY 08/01/2005
PI Name:	Ronca, April Elizabeth Ph.D.		
Project Title:	Hypergravity Effects On The Maternal-Fetal System		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Musculoskeletal Biology (4) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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PI Organization Type:	NASA CENTER	**Phone:**	650.400.6019
Organization Name:	NASA Ames Research Center		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:	November 2019: PI is located at NASA Ames Research Center and remains affiliated with Wake Forest University School of Medicine		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	12/01/2002	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Wade, Charles E Ph.D. (US Army Institute of Surgical Research)		
Grant/Contract No.:	NNA04CK83A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project will advance our understanding of gravity's influence(s) on mammalian pregnancy, parturition and offspring development and their inter-relations. Through a series of integrative experiments, we seek to uncover similarities between prenatal exposure to increased (or hyper-) gravity and other intrauterine events, such as prenatal nutrients and oxygen that 'program', or set the stage for developmental outcomes and adult disease. The fetal programming hypothesis states that prenatal events set into motion a series of physiological alterations in the developmental program that may translate into adult disorders, in particular, cardiovascular disease, diabetes and mental illness. In the current project, Norway rats will be exposed to 2-g or 1.5-g hypergravity to establish continuous centrifugation as a model for fetal programming of adult disease. Non-pregnant females will be adapted to hypergravity prior to mating to eliminate effects of the adaptation process on fetal offspring. We have previously established reduced birth weights of rats exposed to 2-g or 1.5-g centrifugation during prenatal development. We have also observed decreased numbers of labor contractions in dams that underwent pregnancy during 2-g or 1.5-g centrifugation. Behavioral, physiological and molecular methods will be combined to ascertain mechanisms underlying effects of centrifugation on the maternal-fetal system. We will characterize and compare effects of hypergravity on uterine contractile activity during gestation and birth using intrauterine pressure measurement, hormonal and behavioral measures. Studies of fetal offspring at the time of birth will control exposure to uterine contractions by either administering specific numbers of artificially-produced contractions or by completely eliminating exposure to them. Comparisons will be made between offspring that undergo gestation in hypergravity and those exposed to bilateral occlusion of the uterine arteries during late pregnancy, the latter being an established model of fetal programming. Labor contractions will be related to neonatal variables, namely postpartum breathing and oxygenation, suckling and metabolic activity. Molecular analyses will focus on the expression of specific genes that are reactive to sensory simulation, the early, immediate genes, c-fos and c-jun in brain and adrenal of the newborn rats associated with birth. Longitudinal studies of the offspring will focus on adult cardiovascular responses to stress.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The newly emerging field of fetal programming, the idea that perinatal events direct or 'program' adult health and disease states, is uncovering startling insights into relationships between intrauterine events and adult health problems, including cardiovascular disease, diabetes and mental health disorders. Numerous epidemiological and animal studies provide unequivocal evidence for a strong association between low birth weight and adult cardiovascular disease. The goal of the proposed research is to determine whether hypergravity exposure of developing offspring is an appropriate model for fetal programming and can be used to further scientific understanding of fetal programming of adult disease.
Task Progress:	During this funding period, we completed analysis of data collected during a recent centrifuge run. The long-term (sixty-day) centrifuge study involved adapting female rats to centrifugation. Estrus cycling, presence of sperm in the vaginal lavage of females, behavioral aspects of mating, pregnancy rates, birth, and offspring development and maturation were analyzed (Ronca, Rushing, Tou, Wade, & Baer, 2005). Estrous cycles were analyzed by examining vaginal cytology before and for 35 days during centrifugation. Following adaptation to centrifugation, estrous cycle duration over a five-cycle period was similar in centrifuged and non-centrifuged females. Another group of centrifuge-adapted females was time-mated to males that had also been adapted for seven days to centrifugation. Identical numbers of centrifuged and non-centrifuged females conceived, however centrifuged females took four-times longer than controls to achieve conception. Births occurred at the normal gestational length. Pup birth weight and postnatal survival were significantly reduced in centrifuged litters. In conclusion, 2-g centrifugation had no effect on estrus cycle length or the probably of becoming pregnant but delayed conception and diminished pregnancy outcome. These studies are particularly exciting because highly detailed measures of reproductive variables, critical to goals of space exploration and long-term habitation of space. An Important goal of this study, and indeed of the entire proposal, is to establish centrifugation as an experimental model for 'fetal programming'. Fetal programming is the concept that intrauterine events set the stage for adult disease states. In a second study, we tested the hypothesis that rat pups conceived, gestated and born during 2-g centrifugation have significantly reduced birth weights and increased adult body weights as compared to 1-g controls. Offspring were produced by mating young adult male and female rats that were adapted to 2-g centrifugation. Female rats underwent conception, pregnancy and birth at 2-g. Newborn pups in the 2-g condition were removed from the centrifuge and fostered to non-manipulated, newly parturient dams maintained at 1-g. Comparisons were made with 1-g stationary controls, also cross-fostered at birth. As compared to 1-g controls, birth weights of pups gestated and born at 2-g were significantly reduced. Pup body weights were significantly reduced until Postnatal day (P)12. Beginning on P63, body weights of 2-g-gestated offspring exceeded those of 1-g controls by 7-10%. Thus, prenatal rearing at 2-g restricts neonatal growth and increases adult body weight. Collectively, these data support the hypothesis that 2-g centrifugation alters the intrauterine milieu, thereby inducing persistent changes in adult phenotype. Our upcoming studies will focus on mechanisms underlying these astounding lifelong changes.
Bibliography Type:	Description: (Last Updated: 05/15/2025)
Abstracts for Journals and Proceedings	Ronca AE, Baer LA, Rushing L, Wade CE. "Prenatal hypergravity exposure: Evidence for developmental programming. " Symposium presentation FASEB , May-2005
Abstracts for Journals and Proceedings	Ronca AE, Bombick,D, Conway GC, Baer, LA. "Differential patterns of gene expression in newborn rat brain. " Oral Presentation American Psychological Society, Los Angeles, CA , May-2005
Abstracts for Journals and Proceedings	Ronca AE, Bombick D, Conway GC, Baer LA. "Effects of premature birth on c-fos expression in neonatal rats." Poster Society for Neuroscience Meeting , Nov-2005
Abstracts for Journals and Proceedings	Ronca AE, Baer LA, Rushing L, Wade C. "Antenatal programming of adult body weight in male rats induced by centrifugation is associated with elevated leptin and altered body composition. " Oral Presentation ASGSB , Oct-2005

Abstracts for Journals and Proceedings	Ronca AE, Baer LA, Conatser L, McCue R, Saldanha K. "Labor inside and out: Temporal relationships between intrauterine pressure and behavioral expressions of labor in parturient rats. " Poster Developmental Psychobiology , Nov-2005
Articles in Peer-reviewed Journals	Ronca AE, Rushing L, Tou J, Wade CE, Baer LA "Centrifugation effects on estrous cycle, mating success and pregnancy outcome" Journal of Gravitational Physiology , Aug-2005
Articles in Peer-reviewed Journals	Baer LA, Rushing L, Wade CE, Ronca AE. "Prenatal centrifugation: A model for developmental programming of body weight? " Journal of Gravitational Physiology , Aug-2005
Articles in Peer-reviewed Journals	Zakrzewska EI, Maple R, Lintault L, Wade C, Baer L, Ronca A, Plaut K. "Association between gravitational force and tissue metabolism in periparturient rats. " Journal of Gravitational Physiology , Aug-2005
Books/Book Chapters	Alberts JR, Ronca AE. "Development as adaptation: A paradigm for gravitational and space biology" in "Advances in Space Biology and Medicine, Vol. 10, Experimentation with Animal Models in Space" Ed. Sonnenfeld, G. Netherlands: Elsevier, Jan-2005